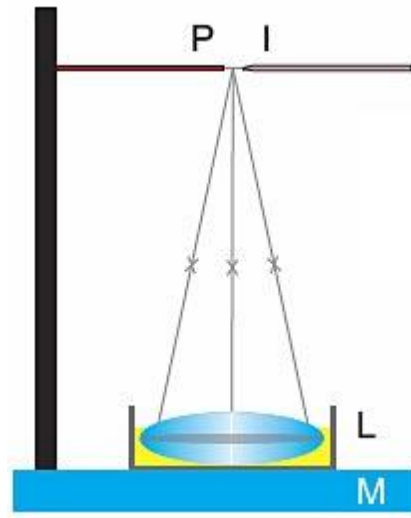


No. of Experiment:

Name of the Experiment: “Determination of the refractive index of liquid by using plane mirror and convex lens”.

Theory:

If by keeping 2 to 3 drops of liquid on a plane mirror and a convex lens of focal length of f_1 is placed on this liquid, then the layer of the liquid between the lens and the plane mirror makes a plano-concave lens of focal length of f_2 . In this situation, the radius of the curvature of lens formed (plano-concave) is equal to the radius of curvature of the convex lens used. The two lenses form a combination of the lenses which acts like a convex lens. Let the focal length of the lens be F .



So, we can write,

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

Here values of F and f_2 are negative.

$$\frac{1}{F} = \frac{1}{f_1} - \frac{1}{f_2}$$

$$\Rightarrow \frac{1}{f_2} = \frac{1}{f_1} - \frac{1}{F} = \frac{F - f_1}{F f_1}$$

$$\Rightarrow \frac{1}{f_2} = \frac{F - f_1}{F f_1}$$

$$f_2 = \frac{F f_1}{F - f_1} \text{ ----- (i)}$$

By measuring the values of f_1 and F and by inserting these values in equation (i) f_2 can be determined.

Here, if refractive index of the liquid with respect to air is μ , radius of the curvature of the spherical surface of the plano-concave lens is r_1 and radius of the plane mirror is r_2 , then the focal length of the plano-concave lens f_2 can be written as –

$$\frac{1}{f_2} = (\mu - 1) \left(\frac{1}{r_1} - \frac{1}{r_2} \right) \text{ ----- (ii)}$$

But if the radius of the plane is infinity, i.e., $r_2 = \infty$, then the equation (ii) can be written as –

$$\frac{1}{f_2} = (\mu - 1) \frac{1}{r_1}$$

$$\Rightarrow f_2(\mu - 1) = r_1$$

$$\Rightarrow (\mu - 1) = \frac{r_1}{f_2}$$

$$\mu = 1 + \frac{r_1}{f_2} \text{ ----- (iii)}$$

List of Apparatus:

1. Experimental convex lens
2. Plane mirror
3. Pin stands
4. Metre scale
5. Slide callipers
6. Spherometer
7. Experimental liquid etc

Procedure:

1. Thickness of the convex lens is determined using a slide callipers. By dividing this thickness t , by 2, the thickness of the upper surface $\frac{t}{2}$ is obtained.
2. By keeping the plane mirror on a table the convex lens is placed on it.
3. Then the objective pin is attached to the stand in such a way as the sharp end remains parallel to the principal axis of lens.
4. Now the objective pin is raised progressively upward. When the parallax error does not remain and real image is formed, then by a metre scale the height h_1 from the mid-point of the upper surface of the lens to the vertex of the pin is measured. Now the focal length of the convex lens is determined by $f_1 = \left(h_1 + \frac{t}{2}\right)$.
5. Removing the convex lens from the plane mirror 2/3 drops of liquid is dropped on the mirror. Again the convex lens is placed on the plane mirror.
6. Again the objective pin is raised upward progressively. When the parallax error does not exist and real image is formed the height h_2 from the mid-point of the upper surface of the lens and the vertex of the pin is measured by the metre scale. Now focal length of the combination of lenses is determined by $F = \left(h_2 + \frac{t}{2}\right)$.
7. The distance between the three legs of the spherometer are measured and average value D is found out. Height from the surface of the plane mirror to the surface of the lens is determined. Then using the formula $r_1 = \left(\frac{D^2}{6h} + \frac{h}{2}\right)$, radius of curvature of the convex lens is found out.

Experimental Data:

Table 1: For the determination of the thickness of the lens.

No. of Obs.	Main scale reading, M (cm)	Vernier scale reading, V	Vernier constant, C (cm)	Value of Vernier reading, $F = V \times C$ (cm)	Total reading, $(M + F)$	Average reading, t (cm)	Reading of the mid-point of the lens, $\frac{t}{2}$ (cm)
1.							
2.							
3.							

Table 2: For the determination of height (h).

No. of Obs.	Which surface	Linear scale reading, M (cm)	Circular scale reading, V	Least count, C (cm)	Value of the circular reading, $F = V \times C$ (cm)	Total reading, $(M + F)$	Average reading, t (cm)	Height, $h = x_2 - x_1$ (cm)
1.	On the plane mirror (x_1)							
2.								
1.	On the lens (x_2)							
2.								

Table 3: For the determination of the values f_1 and F .

No. of Obs.	Height, h_1 (cm)	Average height, h_1 (cm)	Focal length, $f_1 = h_1 + \frac{t}{2}$ (cm)	Height, h_2 (cm)	Average height, h_2 (cm)	Focal length, $F = h_2 + \frac{t}{2}$ (cm)
1.						
2.						
3.						

Determination of the distance, D between the legs of the spherometer:

$$D = \frac{D_1 + D_2 + D_3}{3} = \dots \dots \dots$$

Calculations:

- $r_1 = \left(\frac{D^2}{6h} + \frac{h}{2} \right) = \dots \dots \dots$
- $f_2 = \frac{F f_1}{F - f_1} = \dots \dots \dots$
- $\mu = 1 + \frac{r_1}{f_2} = \dots \dots \dots$

Results:

Refractive index of experimental liquid =

Precautions and Discussion:

1. Appropriate parallax error has been avoided
2. Top of the pin placed along the principal axis of the lens.
3. A small amount of liquid is used on the mirror.
4. Thickness of the lens is correctly measured.